

Research Proposal: Domain-Specific RL Retrieval with Limited Labels

Joint RL + IR project. Sample-efficient reinforcement learning for information retrieval in high-stakes domains (legal, medical) where labeled relevance is scarce.

1. Problem & Motivation

In **legal** and **medical** domains, retrieval quality directly affects decisions and risk: missing relevant precedents or misretrieving clinical evidence can have serious consequences. Three obstacles stand in the way:

- **Scarce labels:** Expert relevance judgments are expensive and slow to obtain.
- **Weak off-the-shelf performance:** General-purpose retrievers are not tuned for domain terminology, document structure, or task-specific success (e.g. citation correctness, downstream QA accuracy).
- **Non-differentiable objectives:** Metrics such as recall@k or end-task accuracy cannot be optimized with standard supervised learning alone.

Evidence of the gap: On **CLERC** (U.S. legal case retrieval), zero-shot IR reaches only **48.3% recall@1000** 1. **LegalBench-RAG** and **CLERC** are the first benchmarks that evaluate the *retrieval* component of RAG in law, stressing minimal relevant snippets rather than document-level retrieval 2, 3. **D3LM** uses positive-unlabeled RL for legal *consultation* (diagnostic questions, court views) but does not optimize the *retrieval* module under limited labels 4. In medicine, **MIRAGE** shows that retrieval design strongly affects LLM accuracy, but RL for *retrieval* with limited labels remains underexplored 5.

We therefore focus on **sample-efficient RL for retrieval** in these high-stakes domains: adapting retrieval models to task-specific goals when relevance labels are limited, using public legal (and optionally medical) benchmarks.

2. Data

We will use public benchmarks to ensure reproducibility and comparability.

Primary (legal):

- **LegalBench-RAG 2:** 6,858 query–answer pairs over 79M+ characters, human-annotated by legal experts; evaluates precise retrieval of minimal relevant text. Includes LegalBench-RAG-mini for fast iteration.
 - arXiv:2408.10343 | GitHub
- **CLERC 1:** U.S. legal case retrieval and retrieval-augmented analysis generation; supports IR (finding citations for analysis) and RAG (generating analyses from citations).
 - arXiv:2406.17186 | ACL Anthology | GitHub

Optional (medical): MIRAGE / MedRAG setting 5: 7,663 questions from five medical QA datasets; we would adopt the same retrieval evaluation protocol if scope allows.

- ACL 2024 Findings

Data splits: We will use official train/validation/test splits where provided, and for sample-efficiency experiments we will subsample training labels (e.g. 10%, 25%, 50%, 100%) with fixed validation/test sets.

3. Methods

Baseline approach: We adapt the **ICLR 2024** framework of 6: policy-gradient RL to optimize pretrained retrieval models for non-differentiable task-specific objectives (recall, MRR, or downstream RAG/QA accuracy).

That work reduces the large action space to binary decisions per query–item pair and uses auxiliary retrievers for exploration; it is directly applicable to legal/medical IR by treating benchmark metrics as rewards.

Our adaptations:

- **Reward:** Reward is defined from retrieval metrics (e.g. recall@k, MRR, nDCG) and/or downstream task accuracy (QA, citation correctness) on the chosen benchmark. When full relevance labels are scarce, we will experiment with bandit-style or preference-based feedback (e.g. pairwise comparisons) to improve sample efficiency.
- **Retriever initialization:** We start from a pretrained retriever (general-purpose or domain-finetuned if available) and adapt it with RL on the target benchmark.
- **Sample-efficiency focus:** We will ablate over the fraction of labeled training data and report performance curves; we may use data augmentation or synthetic queries where consistent with the benchmark license.

Joint IR+RL roles: IR contributes corpus design, relevance criteria, and evaluation metrics; RL contributes policy optimization, reward design, and sample-efficient learning under scarce feedback.

4. Theory Link

- **RL:** The method rests on **policy gradient** optimization of a stochastic policy (retrieval rankings) with respect to a reward that is a function of task-specific metrics. Sample efficiency connects to **exploration–exploitation** (e.g. entropy regularization, auxiliary retrievers as in 6) and to **credit assignment** in multi-step or listwise settings. If we use limited or noisy feedback, **contextual bandits** and **preference-based RL** (e.g. learning from pairwise comparisons) provide a theoretical basis for learning from partial information.
 - **IR:** Relevance is formalized in **probability ranking principle** and **learning-to-rank** (LTR); our reward functions align with standard IR metrics (recall@k, MRR, nDCG). The project bridges LTR (supervised from relevance labels) and **reinforcement learning for ranking**, where the objective is non-differentiable or labels are scarce—a setting that has been studied in general retrieval 6 but not yet systematically in legal/medical domains.
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5. Evaluation Plan

Primary metrics: Retrieval quality on held-out test sets: **recall@k** ($k = 1, 5, 10, 100$), **MRR**, **nDCG**. Where the benchmark defines a downstream task (QA, citation), we also report **downstream accuracy** (e.g. exact match, F1, or benchmark-specific metric).

Baselines: (1) Zero-shot pretrained retriever; (2) Supervised fine-tuning on full (or same subsampled) labels when applicable; (3) Existing legal/medical retrieval or RAG results reported on LegalBench-RAG and CLERC.

Ablations: (1) **Sample efficiency:** performance vs. fraction of training labels (e.g. 10%, 25%, 50%, 100%); (2) **Reward choice:** retrieval-only vs. retrieval + downstream reward; (3) **Initialization:** general vs. domain-adapted retriever.

Protocol: Fixed random seeds and data splits; report mean and standard deviation over multiple runs where feasible. We will release code and configs to reproduce results.

6. Risks

Risk	Mitigation
Too few labels to learn reliably	Use LegalBench-RAG-mini and CLERC for iteration; focus on sample-efficiency ablations as a contribution; consider preference/bandit feedback.
Reward design (e.g. gaming recall)	Align reward with downstream task where possible; monitor for degenerate rankings; validate on held-out downstream metrics.
Compute and runtime	Start with LegalBench-RAG-mini and smaller retriever; scale to full benchmark once pipeline is stable.
Domain shift (legal vs. medical)	Primary scope is legal (LegalBench-RAG, CLERC); medical extension only if time permits, with clear scope limit.
Reproducibility of baselines	Use public code and checkpoints where available; document all data splits and hyperparameters.

7. Work Plan

Phase	Activities	Output
1. Setup	Acquire LegalBench-RAG and CLERC; set up retriever baseline and evaluation pipeline; reproduce zero-shot and (if available) supervised numbers.	Data loaded; baseline results on dev set.
2. RL adaptation	Implement or adapt policy-gradient RL from 6; define reward from benchmark metrics; train on LegalBench-RAG-mini then full (or CLERC).	RL-adapted retriever; initial results.
3. Sample efficiency & ablations	Subsample training labels (10%, 25%, 50%, 100%); run reward and initialization ablations; collect sample-efficiency curves.	Ablation tables and figures.
4. Write-up & release	Finalize experiments; write report/paper; release code and configs.	Draft paper; public code.

Medical benchmark (if in scope) can be added in parallel or as a short extension after Phase 3.

References (with Links)

- 1 CLERC: A Dataset for U.S. Legal Case Retrieval and Retrieval-Augmented Analysis Generation.** arXiv:2406.17186. NAACL 2025 Findings. [arXiv](#) | [ACL](#) | [GitHub](#)
- 2 LegalBench-RAG: A Benchmark for Retrieval-Augmented Generation in the Legal Domain.** arXiv:2408.10343. [arXiv](#) | [GitHub](#)
- 3 A Reasoning-Focused Legal Retrieval Benchmark.** [RegLab](#)
- 4 Knowledge-Infused Legal Wisdom: Navigating LLM Consultation through the Lens of Diagnostics and Positive-Unlabeled Reinforcement Learning** Wu et al. Findings of ACL 2024. [ACL](#)
- 5 Benchmarking Retrieval-Augmented Generation for Medicine (MIRAGE).** Xiong et al. Findings of ACL 2024. [ACL](#)
- 6 Adapting Retrieval Models to Task-Specific Goals using Reinforcement Learning.** ICLR 2024. [OpenReview](#) | [arXiv](#)